

Scientific Computing & Advanced Programming Lab

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Lecture-1



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Welcome!

-
- ▶ Slides will be available at
"https://sauravsrar.github.io/MA-5105-Aug-26/"
 - ▶ Codes corresponding to each lecture will be available
at the same place.



Prerequisites and Demands

Prerequisites

- ▶ Some python programming experience
- ▶ Basic linear algebra and calculus



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Demands

- ▶ honesty: don't cheat yourself !



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- ▶ clocking numerous hours per week in writing / testing codes.



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- ▶ Basic linear algebra and calculus

Demands

- ▶ honesty: don't cheat yourself !
- ▶ clocking numerous hours per week in writing / testing codes.
- ▶ interact with me and the teaching assistant



Course Policies

Collaboration

- ▶ struggle first



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- ▶ discuss specific doubts over the macroscopic problem



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- ▶ don't use to solve the problem



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- ▶ don't use to solve the problem
- ▶ may be only ask for clarifications



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Assignments

- ▶ No assignments



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Assignments

- ▶ No assignments
- ▶ A lot of problem sets



Grading

- ▶ 4 surprise quizzes (fairly easy), 10 Marks each.
- ▶ 1 viva exam, 20 Marks.
- ▶ Class participation: 20 Marks.
- ▶ A final project worth 20 Marks.



Computing

- What do computers do ?



Computing

- ▶ What do computers do ?
- ▶ How do they compute ?



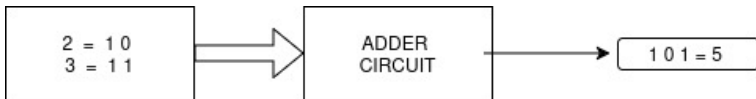
Computing

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 1. Data movement: read / store
 2. logical : Compare / And / Or
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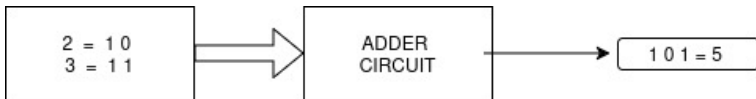
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- ▶ Example-1: Calculate $2 + 3$:



- ▶ By itself not very impressive.



Computing

Example-2:

1. Read marks of next student;
2. Add score;
3. Compute Avg;
4. Store result;



Computing

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Computing

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Computing

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1. Read marks of next student;
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 3. Compute Avg;
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 5. Repeat steps for every student;
 6. Print the final grade list.
- ▶ the computer doesn't learn any new operation for each of the instructions
 - ▶ each operation is broken down to basic operation blocks, performed in some sequence



What is programming?

- ▶ Programming is the process of organising basic operations into simple applications which combine to solve a much larger problem.



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- ▶ can compute anything using 6 primitives (Turing)
- ▶ programming language provides a set of primitive operations
- ▶ create "expressions" which are complex but legal combination of primitives
- ▶ write down a sequence of declaration / evaluation usually to meet some end goal.



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- ▶ create "expressions" which are complex but legal combination of primitives
- ▶ write down a sequence of declaration / evaluation usually to meet some end goal.
- ▶ Complex software is built by combining numerous of these tiny, simple instructions into a carefully organised sequence.



What is a program?



What is a program?

Program

A program is a sequence of definitions and commands

- ▶ definitions **evaluated**
- ▶ commands **executed**



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Scripts/Shells

- ▶ can be typed directly in a shell
- ▶ typically stored in a file



Install and use Python through IDLE (Ubuntu)

▶ `python3 --version`



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- ▶ `idle-python3.12` (**start idle**)



Where things may go wrong

Syntactic errors:

- ▶ common
- ▶ easily caught by compilers
- ▶ English:
 - “sat cat mat” → not syntactically valid
 - “cat sat on the mat” → syntactically valid
- ▶ Programming Language:
 - “hi” 5 → not syntactically valid
 - $3.2 * 5$ → syntactically valid



Where things may go wrong

Static semantic errors:

- ▶ syntactically valid strings
- ▶ no semantic errors
 1. program crashes, stops running
 2. program runs forever
 3. program gives an answer but different than expected



Content

Objects

Branching and Iterations



OBJECTS

- ▶ programs manipulate data objects



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- ▶ objects have a type
 - ▶ Saurav is a human so he can breathe, sleep, etc.
 - ▶ An uncle is a relative, so he can give career advice, discuss politics, ask about your grades.
 - ▶ Mango is a fruit, so it can be ...



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- ▶ Objects are:
 1. Scalar (cannot be subdivided)
 2. non-scalar (have internal structure that can be accessed)



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Scalar objects

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- ▶ `NoneType` - **special** and has one value, None



Scalar objects

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- ▶ `float` - represents **real numbers**. e.g. 3.14159265
- ▶ `bool` - represents **real Boolean values** True or False
- ▶ `NoneType` - **special** and has one value, None
- ▶ use `type (..)` to see the type of an object



Code1

```
# Scalar Data type
```

```
x = 10           # int  
y = 3.14        # float  
z = 2 + 5j      # complex  
flag = True     # bool  
type(x)  
print(type(x))
```



Scalar objects

Type conversion (cast)

- ▶ can convert object of one type to another
- ▶ `float (3)` converts integer 3 to float 3.0
- ▶ `int (3.9)` truncates float 3.9 to integer 3



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Print to console

- ▶ `3 + 2`



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Print to console

- ▶ `3 + 2`
- ▶ `print (3 + 2)`



Variables

Binding Variables and values

```
► interest = 1000.0 * 9.8 * 6 / 100.0
```



Variables

Binding Variables and values

- ▶ `interest = 1000.0 * 9.8 * 6 / 100.0`
- ▶ `roi = 9.8`
- ▶ `prin = 1000.0`
- ▶ `time = 6`



Variables

Binding Variables and values

- ▶ `interest = 1000.0 * 9.8 * 6 / 100.0`
- ▶ `roi = 9.8`
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- ▶ `interest = prin * roi * time / 100.0`



Variables

Binding Variables and values

- ▶ `interest = 1000.0 * 9.8 * 6 / 100.0`
- ▶ `roi = 9.8`
- ▶ `prin = 1000.0`
- ▶ `time = 6`
- ▶ `interest = prin * roi * time / 100.0`
- ▶ can modify interest by modifying any of the basic components.



Code3

```
# definitions <- This is a comments
```

```
pi = 3.14159
```

```
radius = 2.2
```

```
area = pi * radius # what is this?
```

```
print('area = ', area)
```

```
# use comments to help others understand what y
```

```
radius = radius + 1
```

```
print(area)           # area doesn't change
```

```
# Recompute
```

```
area = pi*(radius**2)
```

```
print(area)           # are is updated
```



Math Functions and Number Methods

1. `round()`, for rounding numbers to some number of decimal places

```
▶ >>> round(2.3)
>>> round(3.14159, 3)
>>> round(2.65, 1.4)
```



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- ▶

```
>>> round(2.3)
>>> round(3.14159, 3)
>>> round(2.65, 1.4)
```

2. `abs()`, for getting the absolute value of a number

- ▶

```
>>> abs(3)
>>> abs(-5.0)
```

3. `pow()`, for raising a number to some power

- ▶

```
>>> pow(2, 3)
>>> pow(2, -2)
>>> pow(2, 3, 2)
```



Strings

- ▶ Many programmers deal with text on a daily basis
- ▶ example
 - ▶ web developers work with text that gets input from web forms
 - ▶ Data scientists process text to extract data and perform things like sentiment analysis, to classify opinions in a body of text.



Strings

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 - ▶ web developers work with text that gets input from web forms
 - ▶ Data scientists process text to extract data and perform things like sentiment analysis, to classify opinions in a body of text.
- ▶ Collections of text in Python are called strings



Strings

- ▶ letters, special characters, spaces, digits



Strings

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- ▶ enclose in

`hi = 'hello there' or`

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Strings

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```
hi = 'hello there' or
```

```
hi = "hello there" or
```

- ▶ **concatenate** strings

```
name = "saurav"
```

```
greet = hi + name
```

```
greeting = hi + " " + name
```

- ▶ **String Indexing**

```
>>> flavor = "apple pie"
```

```
>>> flavor[1]
```

```
'p'
```

- ▶ In Python—and most other programming languages—counting always starts at zero.



Strings

- ▶ To get the character at the beginning of a string, you need to access the character at position 0:

```
>>> flavor[0]
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- ▶ **String Slicing**

- ▶ Suppose you need the string containing just the first three letters of the string "apple pie".

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>>> first_three_letters = flavor[0] +  
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>>> first_three_letters  
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Strings

Strings Are Immutable

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```
>>> word = "goal"
```

```
>>> word[0] = "f"
```



Strings

Strings Are Immutable

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```
>>> word = "goal"
>>> word[0] = "f"
```
- ▶ If we want to alter a string, we must create an entirely new string

Converting Strings to Numbers

- ▶ `int()` stands for integer and converts objects into whole numbers,
- ▶ `float()` stands for floating-point number and converts objects into numbers with decimal points.



Strings

Converting Numbers to Strings

- ▶ Sometimes you need to convert a number to a string.
- ▶

```
>>> num_pancakes = 10  
>>> "I am going to eat " + num_pancakes  
+ " pancakes."
```



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Converting Numbers to Strings

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>>> num_pancakes = 10  
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>>> num_pancakes = 10  
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+ " pancakes."
```
- ▶ Since `num_pancakes` is a number, Python can't concatenate it with the string "I'm going to eat".
- ▶ To build the string, you need to convert `num_pancakes` to a string using `str()`:

```
>>> num_pancakes = 10  
>>> "I am going to eat " +  
str(num_pancakes) + " pancakes."
```



Input / Output

► output

```
>>> x = 1
```

```
>>> print(3)
```



Input / Output

► output

```
>>> x = 1  
>>> print(3) >>> print(5*x)
```

Interact With User Input

► input

```
>>> text = input("Type anything... ")  
>>> print (text)
```



Input / Output

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>>> x = 1  
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Interact With User Input

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```

► Working With Strings and Number:

user input using the `input()` function, the result is always a string.



Content

Objects

Branching and Iterations



Comments

```
import math

principal = 500000
interest = 7.5
duration = 20

r = interest / 12 / 100
n = duration * 12

if r == 0:
    payment = principal / n
else:
    payment = principal * r * (1 + r) ** n / ((1 + r) ** n - 1)

print("Monthly EMI = ", payment, " Ruppees")
```



Comments

A person can understand the syntax, but they may wonder:



Comments

A person can understand the syntax, but they may wonder:

- ▶ Why is the interest divided by 12 and then by 100?
- ▶ Where did this complicated formula come from?
- ▶ Why is there a special case when $r == 0$?

The code works, but the intent is not obvious.



Comments

```
import math

# Loan details
principal = 500000      # Loan amount in rupees
interest = 7.5          # Annual interest rate (in percent)
duration = 20           # Loan repayment period (in years)

# Convert the annual interest rate into a monthly decimal rate,
# as the EMI formula works with monthly interest.
r = interest / 12 / 100

# Total number of monthly installments.
n = duration * 12

# If there is no interest, each installment is simply an equal
# share of the principal.
if r == 0:
    payment = principal / n
else:
    # Compute the Equated Monthly Installment (EMI) using the
    # standard loan amortization formula used by banks.
    payment = principal * r * (1 + r) ** n / ((1 + r) ** n - 1)

# Display the monthly payment amount.
print("Monthly EMI =", payment, "Rupees")
```



Comments

What the comments add

- ▶ Why the interest rate is converted.
- ▶ What n represents.
- ▶ Why there is a special case for zero interest.
- ▶ Where the formula comes from (the standard EMI formula).



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What the comments add

- ▶ Why the interest rate is converted.
- ▶ What n represents.
- ▶ Why there is a special case for zero interest.
- ▶ Where the formula comes from (the standard EMI formula).
- ▶ What each input variable means.



Comments

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# Print "Hello, world"  
print("Hello, world")
```



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- ▶ to explain why something is done a certain way



Comments

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- ▶ No comment is needed in this example.
- ▶ best used to clarify code that may not be easy to understand
- ▶ to explain why something is done a certain way
- ▶ comments should be used sparingly
- ▶ Use comments only when they add value



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- ▶ These are all built-in functions because they come built into the Python language itself.
- ▶ You can also create user-defined functions that perform specific tasks.
- ▶ Functions break code into smaller chunks, and are great for tasks that a program uses repeatedly



Functions Are Values

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- ▶ Like any other variable, you can assign any value you want to len:

```
>>> len= "I'm not the len you're  
looking for."  
>>> len
```



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looking for."  
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```

- ▶ Now `len` has a string value
- ▶ The variable name `len` is a keyword in Python, and even though you can change it's value, it's usually a bad idea to do so.



Functions Are Values

- ▶ Changing the value of `len` can make your code confusing



Functions Are Values

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- ▶ because it's easy to mistake the new `len` for the built-in function.



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Functions Are Values

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- ▶ because it's easy to mistake the new `len` for the built-in function.
- ▶ If you typed in the previous code examples, you no longer have access to the built-in `len` function in IDLE.
- ▶ You can get it back with the following code:

```
>>> del len
```
- ▶ The `del` keyword is used to un-assign a variable from a value.
- ▶ `del` stands for delete, but it doesn't delete the value. Instead, it detaches the name from the value and deletes the name.



How Python Executes Functions

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```
>>> len()
```

- ▶ Python raises a `TypeError` when `len()` is called because `len()` expects an argument.
- ▶ An argument is a value that gets passed to the function as input.
- ▶ Some functions can be called with no arguments, and some can take as many arguments as you like.
- ▶ When a function is done executing, it returns a value as output.



How Python Executes Functions

The process for executing a function can be summarized in three steps:

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1. The function is called, and any arguments are passed to the function as input.
2. The function executes, and some action is performed with the arguments.
3. The function returns, and the original function call is replaced with the return value.

```
>>> num_letters = len("four")
```

- ▶ When a function changes or affects something external to the function itself, it is said to have a side effect.

```
>>> return_value= print("What do I  
return?")
```



User define Functions

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Every function has two parts:

1. The **function signature** defines the name of the function and any inputs it expects.



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Every function has two parts:

1. The **function signature** defines the name of the function and any inputs it expects.
2. The **function body** contains the code that runs every time the function is used.



Multiplication function

- ▶ write a function that takes two numbers as input and returns their product.

```
def multiply(x, y):    # Function
signature
    # Function body
    product = x * y
    return product
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- ▶ The first line of code in a function is called the function signature.
- ▶ It always starts with the `def` keyword, which is short for “define.”



Function Signature

```
def multiply(x, y):
```

The function signature has four parts:

- ▶ The `def` keyword



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 - ▶ A colon `(:)` at the end of the line
-
- ▶ When Python reads a line beginning with the `def` keyword, it creates a new function.
 - ▶ The function is assigned to a variable with the same name as the function name.



Functions

- ▶ a function name can only contain numbers, letters, and underscores, and must not begin with a number.



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# Function body  
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```

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- ▶ It starts with the `return` keyword and is followed by the variable `product`.



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- ▶ It tells Python which statements belong together as a block.

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def multiply(x, y):  
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print("Where am I?") # Not in the function body.
```



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```

- ▶ In Python, indentation is mandatory, not just a matter of style.



Indentation in Python

Incorrect Indentation

- ▶ If `print()` is indented, then it becomes a part of the function body even if there is a blank line between `print()` and the previous line



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marks = 45
```

```
if marks >= 50:  
    print("Marks = ", marks)  
print("Pass")
```



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```
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- ▶ There is one rule every line must be indented by the same number of spaces.



Function Return

- ▶ Once Python executes a return statement, the function stops running and returns the value.



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def multiply(x, y):  
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```

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print("Where am I?") # Not in the function body.
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- ▶ If any code appears below the return statement that is indented so as to be part of the function body, it will never run



Calling a User-defined Function

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- ▶ Type the function name followed by a list of arguments in between parentheses.



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```

- ▶

```
num = multiply(2, 4)
```



```
print(num)
```

```
def multiply(x, y):  
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    return product
```



EMI Calculator

Write a function to generalise the EMI Calculator and compute the interest for one particular case.

```
import math

# Loan details
principal = 500000          # Loan amount in rupees
interest = 7.5              # Annual interest rate (in percent)
duration = 20               # Loan repayment period (in years)

# Convert the annual interest rate into a monthly decimal rate,
# as the EMI formula works with monthly interest.
r = interest / 12 / 100

# Total number of monthly installments.
n = duration * 12

# If there is no interest, each installment is simply an equal
# share of the principal.
if r == 0:
    payment = principal / n
else:
    # Compute the Equated Monthly Installment (EMI) using the
    # standard loan amortization formula used by banks.
    payment = principal * r * (1 + r) ** n / ((1 + r) ** n - 1)

# Display the monthly payment amount.
print("Monthly EMI =", payment, "Rupees")
```



Functions With No Return Statement

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def greet(name):  
    print(f"Hello, {name}")
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def greet(name):  
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- ▶ Even though `greet()` has no return statement, it still returns a value:

```
return_value= greet("Dave")
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- ▶ Even though `greet()` has no return statement, it still returns a value:

```
return_value= greet("Dave")
```
- ▶ Notice also that the string `"Hello, Dave!"` is printed even when the result of `greet("Dave")` is assigned to a variable.
- ▶ That's because the call to `print()` inside of the `greet()` function body produces the side effect of always printing to the console.



Documenting Your Functions

```
► >>> help(len)
Help on built-in function len in module
builtins:
len(obj, /)
Return the number of items in a
container.
```



Documenting Your Functions

- ▶ `>>> help(len)`
Help on built-in function len in module builtins:
`len(obj, /)`
Return the number of items in a container.
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Documenting Your Functions

- ▶ `>>> help(len)`
Help on built-in function len in module builtins:
`len(obj, /)`
Return the number of items in a container.
- ▶ When you pass a variable name or function name to `help()`, it displays some useful information about it.
- ▶ Let's see what happens when you call `help()` on the `multiply()` function:
- ▶ `help()` displays the function signature, but there isn't any information about what the function does.



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Documenting Your Functions

- ▶ To better document `multiply()`, we need to provide a `docstring`.
- ▶ A `docstring` is a triple-quoted string literal placed at the top of the function body.
- ▶ Docstrings are used to document what a function does and what kinds of parameters it expects.
- ▶ Here's what `multiply()` looks like with a docstring added to it:

```
def multiply(x, y):  
    """Return the product of 2 numbers x and y."""  
    product = x * y  
    return product
```



Run in Circles

- ▶ A loop is a block of code that gets repeated over and over again either a specified number of times or until some condition is met.



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`while` loops repeat a section of code while some condition is true. There are two parts to every `while` loop:

1. The `while` statement starts with the `while` keyword, followed by a `test condition`, and ends with a `colon (:)`.
2. The `loop body` contains the code that gets repeated at each step of the loop. Each line is indented.



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- ▶ drop the last statement `n = n + 1`
- ▶ If you aren't careful, you can create an **infinite loop**.
- ▶ This happens when the test condition is always true.
- ▶ An infinite loop never terminates.



while Infinite loops

- ▶ Infinite loops aren't inherently bad.



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- ▶ For example, code that interacts with hardware may use an infinite loop to constantly check whether or not a button or switch has been activated.
- ▶ If you run a program that enters an infinite loop, you can force Python to quit by pressing Ctrl+C.



while Infinite loops

- ▶ Infinite loops aren't inherently bad.
- ▶ Sometimes they are exactly the kind of loop you need.
- ▶ For example, code that interacts with hardware may use an infinite loop to constantly check whether or not a button or switch has been activated.
- ▶ If you run a program that enters an infinite loop, you can force Python to quit by pressing Ctrl+C.
- ▶ Python stops running the program and raises a Keyboard Interrupt error:

```
Traceback (most recent call last):
```

```
File "<pyshell#8>", line 2, in <module>  
    print(n)
```

KeyboardInterrupt



Example: Estimating e Using a `while` Loop

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$$



Example: Estimating e Using a `while` Loop

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Problem Approximate the value of e by successively adding terms of the series.



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Problem Approximate the value of e by successively adding terms of the series. Continue adding terms until the next term satisfies $\frac{1}{n!} < 10^{-8}$.

- This is a common stopping criterion in numerical mathematics.



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Problem Approximate the value of e by successively adding terms of the series. Continue adding terms until the next term satisfies $\frac{1}{n!} < 10^{-8}$.

- ▶ This is a common stopping criterion in numerical mathematics.

Why is a `while` loop appropriate?

- ▶ The number of terms required is *not known in advance*.
- ▶ The computation stops only when a desired accuracy is achieved.



Python Implementation

```
n = 0
term = 1.0
total = 0.0

while term > 1e-8:
    total += term
    n += 1
    term = term / n

print("Approximation of e =", total)
print("Terms used =", n)
```



Finding a Root Using a `while` Loop

$$f(x) := x^3 - x - 2 = 0.$$

$$f(1) = -2, \quad f(2) = 4,$$

so there is a root in the interval $[1, 2]$.



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so there is a root in the interval $[1, 2]$.

Problem

Use the *bisection method* to approximate the root.



Finding a Root Using a `while` Loop

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$$f(1) = -2, \quad f(2) = 4,$$

so there is a root in the interval $[1, 2]$.

Problem

Use the *bisection method* to approximate the root. Continue halving the interval until its length is less than 10^{-6} .

Why use a `while` loop?

- ▶ The number of iterations is not known beforehand.
- ▶ The algorithm stops only when the desired accuracy is achieved.



Python Implementation

```
def f(x):  
    return x**3 - x - 2  
  
a = 1.0  
b = 2.0  
tol = 1e-6  
  
while (b - a) > tol:  
    c = (a + b) / 2  
  
    if f(a) * f(c) < 0:  
        b = c  
    else:  
        a = c  
  
root = (a + b) / 2  
  
print("Approximate root =", root)
```



The `for` Loop

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Like `while` the `for` loop has two main parts:

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Like `while` the `for` loop has two main parts:

1. The `for` statement begins with the `for` keyword, followed by a membership expression, and ends in a colon (`:`).
2. The `loop body` contains the code to be executed at each step of the loop, and is indented.



The for Loop

Type:

```
for letter in "Python":  
    print(letter)
```



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How to do the same with `while` loop



The for Loop

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for letter in "Python":  
    print(letter)
```

How to do the same with `while` loop

```
word= "Python"  
index= 0  
while index < len(word):  
    print(word[index])  
    index= index + 1
```



Numerical Integration Using a `for` Loop

$$I = \int_0^1 x^2 dx.$$



Numerical Integration Using a `for` Loop

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- ▶ approximate the integral using the **Left Rectangle Rule**.



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$$I = \int_0^1 x^2 dx.$$

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- ▶ Divide the interval $[0, 1]$ into N equal subintervals of width $\Delta x = \frac{1}{N}$



Numerical Integration Using a `for` Loop

$$I = \int_0^1 x^2 dx.$$

- ▶ approximate the integral using the **Left Rectangle Rule**.
- ▶ Divide the interval $[0, 1]$ into N equal subintervals of width $\Delta x = \frac{1}{N}$

$$I \approx \sum_{i=0}^{N-1} f(x_i) \Delta x, \quad x_i = i \Delta x, \quad f(x) = x^2.$$

Task: Approximate the integral using $N = 100$ subintervals.

Why use a `for` loop?

- ▶ The number of rectangles (N) is known in advance.
- ▶ The same computation is repeated exactly N times.



Python Implementation



Nested for Loops

A **nested loop** is a loop inside another loop.

General Syntax

```
for i in range(...):  
    # Statements in the outer loop  
  
    for j in range(...):  
        # Statements in the inner loop  
  
    # More statements in the outer loop
```

How it works

- ▶ The outer loop executes once.
- ▶ For each iteration of the outer loop, the inner loop

~~runs from start to finish~~



CONTROL FLOW - BRANCHING

